Calculus with Parametric and Polar Curves

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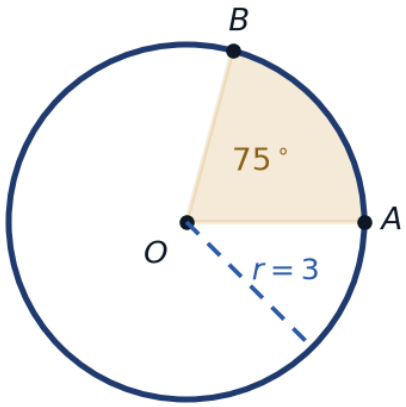
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Parametric derivatives and arc length

For the curve $x = t^2$, $y = t^3$, find $\frac{dy}{dx}$ in terms of t.

2 [Foundational] • Target: Parametric derivatives and arc length

Find the arc length of the curve $x = 3t^2$, $y = 2t^3$ for $0 \le t \le 2$.



3 [Foundational] • Target: Polar area

Find the area enclosed by one loop of the rose $r = 3\sin 2\theta$ (for $0 \le \theta \le \pi/2$).

4 [Foundational] • Target: Polar area

Find the area enclosed by the cardioid $r = 2 + 2\cos\theta$.

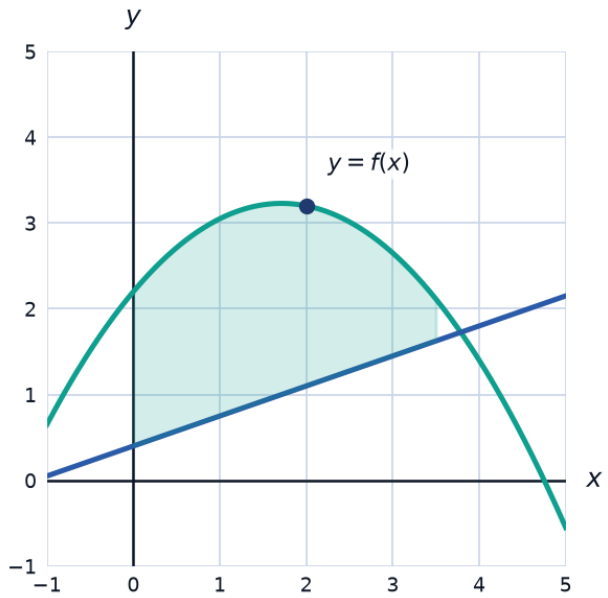
5 [Foundational] • Target: Second derivatives and concavity for parametric curves

For $x = t^2$, $y = t^3 - 3t$, find $\frac{d^2y}{dx^2}$ in terms of t, using $\frac{d^2y}{dx^2} = \left[\frac{d}{dt}\left(\frac{dy}{dx}\right)\right] \div \frac{dx}{dt}$.

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6 [Foundational] • Target: Area between polar curves

Find the area inside the circle $r = 3$ and outside the circle $r = 2$ (an annular region).



7 [Foundational] • Target: Horizontal and vertical tangents on parametric curves

For $x = t^2 - 4$, $y = t^3 - 3t$, find the value(s) of t where the tangent line is horizontal ($\frac{dy}{dt} = 0$, $\frac{dx}{dt} \neq 0$).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Horizontal and vertical tangents on parametric curves

For the same curve, find the value(s) of t where the tangent line is vertical ($\frac{dx}{dt} = 0$, $\frac{dy}{dt} \neq 0$).

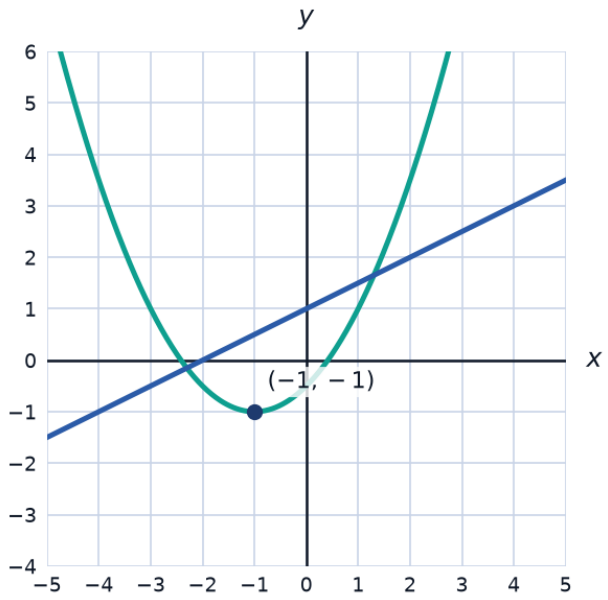
9 [Intermediate] • Target: Converting between polar and Cartesian for calculus

Find the slope of the tangent line to $r = 1 + \sin\theta$ at $\theta = 0$, using $x = r\cos\theta$, $y = r\sin\theta$ as parametric equations in θ.

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10 [Intermediate] • Target: Speed and length for parametric motion

A particle moves according to $x(t) = t^2$, $y(t) = 4t$. Find its speed at $t = 3$.



11 [Intermediate] • Target: Speed and length for parametric motion

Find the arc length of $x = \cos^3 t$, $y = \sin^3 t$ for $0 \le t \le \frac{\pi}{2}$ (an astroid arc), using $\frac{dx}{dt} = -3\cos^2 t \sin t$ and $\frac{dy}{dt} = 3\sin^2 t \cos t$.

12 [Intermediate] • Target: Slope and concavity for polar-derived curves

Find $\frac{dy}{dx}$ for the polar curve $r = 2\theta$ at $\theta = \frac{\pi}{2}$, using $x = r\cos\theta$, $y = r\sin\theta$.

13 [Intermediate] • Target: Slope and concavity for polar-derived curves

For $x = t^3 - 3t$, $y = t^2$, find the points where the tangent is horizontal.

14 [Intermediate] • Target: Arc length of a polar curve

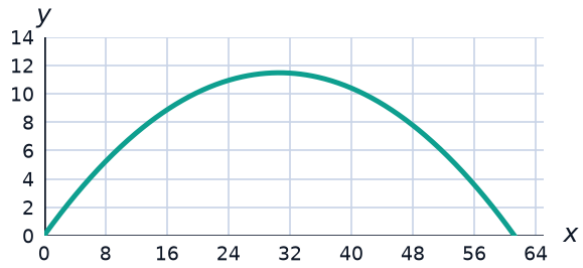
Set up (do not evaluate) the arc length integral for the circle $r = 4$ over $0 \le \theta \le 2\pi$, using $L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$, and verify the setup is consistent with the known circumference formula.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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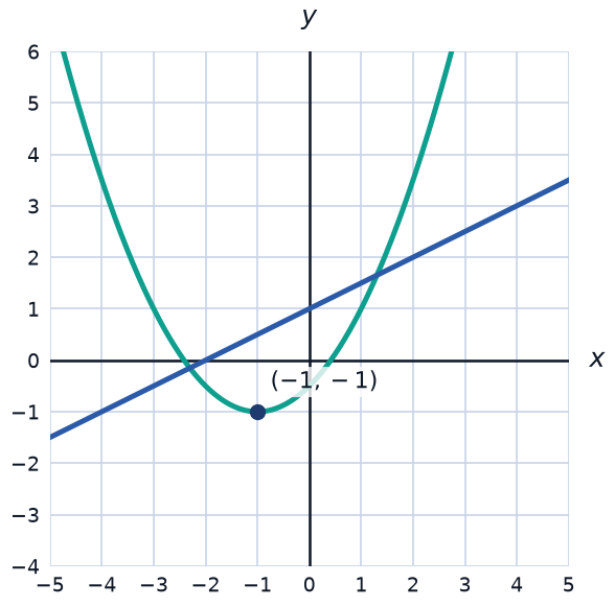
15 [Challenging] • Target: Applied parametric motion • Cross-Lesson: 6.01 Related Rates

A projectile's position is given by $x(t) = 20t$, $y(t) = 15t - 4.9t^2$ (meters, seconds). Find the projectile's velocity components $\frac{dx}{dt}$ and $\frac{dy}{dt}$, and find the speed at $t = 1$.



16 [Challenging] • Target: Applied parametric motion • Cross-Lesson: 6.02 Optimization and Curve Sketching

For the same projectile, find the time at which the vertical velocity is 0 (the peak of the trajectory).



17 [Challenging] • Target: Polar curves and symmetry • Cross-Lesson: 7.01 Integration by Substitution

The cardioid $r = 1 + \cos\theta$ is symmetric about the polar axis. Use this symmetry to explain why the total area enclosed can be computed as $2\times$ the area for $0 \le \theta \le \pi$, and evaluate the total area.

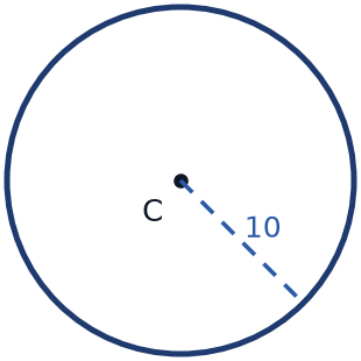
18 [Challenging] • Target: Polar curves and symmetry • Cross-Lesson: 7.02 Integration by Parts

For the parametric curve $x = e^t\cos t$, $y = e^t\sin t$, find $\frac{dy}{dx}$ at $t = 0$, using $\frac{dx}{dt} = e^t(\cos t - \sin t)$ and $\frac{dy}{dt} = e^t(\sin t + \cos t)$.

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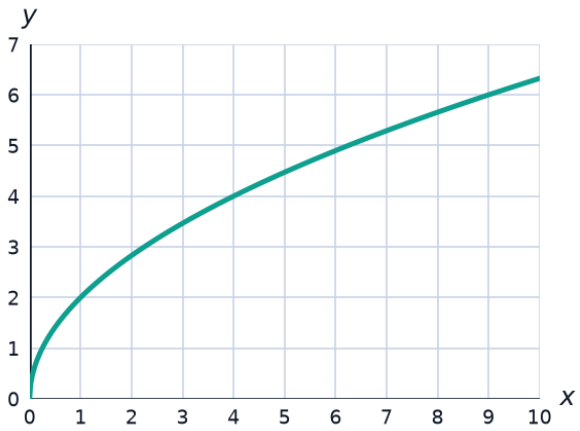
19 [Challenging] • Target: Polar curves and symmetry • Cross-Lesson: 8.01 Area Between Curves

A Ferris wheel of radius 10 m, centered 12 m above the ground, rotates so that a rider's position is $x(t) = 10\cos t$, $y(t) = 12 + 10\sin t$, where t is measured in radians and represents elapsed angle since the rider was at the rightmost point. Find the rider's height above the ground when $t = \frac{\pi}{3}$, and find the vertical velocity $\frac{dy}{dt}$ at that instant (assuming t increases at a constant rate of 1 rad per unit time).



20 [Challenging] • Target: Total distance traveled along a parametric path • Cross-Lesson: 8.02 Volumes of Revolution

A robot on a factory floor follows the path $x(t) = t^2$, $y(t) = 2t$ for $0 \le t \le 3$ (meters, seconds). Set up the arc length integral for the total distance traveled, and evaluate it exactly using the substitution $u = t^2 + 1$ where appropriate.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Derivatives of Parametric Curves

Given $x(t) = t^3 - 3t$ and $y(t) = t^2 + 1$, find $\frac{dy}{dx}$ in terms of t.

a. $\frac{2t}{3t^2 - 3}$

b. $\frac{3t^2 - 3}{2t}$

c. $\frac{2t}{3t^2}$

d. $\frac{2}{3t}$

22 [Foundational] • Target: Derivatives of Parametric Curves

Find the equation of the tangent line to the parametric curve $x = 2\cos t$, $y = 3\sin t$ at $t = \pi/4$.

a. $y - \frac{3\sqrt{2}}{2} = -\frac{3}{2}(x - \sqrt{2})$

b. $y - \sqrt{2} = -\frac{2}{3}(x - \frac{3\sqrt{2}}{2})$

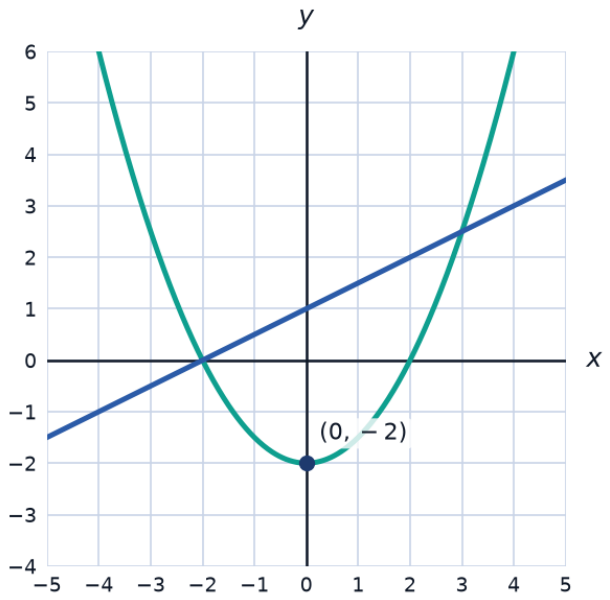
c. $y = -\frac{3}{2}x$

d. $y - 3 = -(x - 2)$

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23 [Foundational] • Target: Derivatives of Parametric Curves

State the formula for the second derivative $\frac{d^2y}{dx^2}$ of a parametric curve.



a. $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left[\frac{dy}{dx}\right]}{\frac{dx}{dt}}$

b. $\frac{d^2y}{dx^2} = \frac{y''(t)}{x''(t)}$

c. $\frac{d^2y}{dx^2} = \frac{y''(t)}{[x'(t)]^2}$

d. $\frac{d^2y}{dx^2} = \frac{d}{dt}\left[\frac{dy}{dx}\right]$

24 [Foundational] • Target: Arc Length of Parametric Curves

State the formula for the arc length of a parametric curve from $t = a$ to $t = b$.

a. $L = \int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2}\, dt$

b. $L = \int_a^b (x'(t) + y'(t))\, dt$

c. $L = \int_a^b \sqrt{1 + [y'(t)]^2}\, dt$

d. $L = \int_a^b \sqrt{[x(t)]^2 + [y(t)]^2}\, dt$

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25 [Foundational] • Target: Arc Length of Parametric Curves

Find the arc length of $x = 3\cos t, y = 3\sin t$ for $0 \le t \le 2\pi$.

a. 6π

b. 3π

c. 9π

d. 12π

26 [Foundational] • Target: Arc Length of Parametric Curves

Find the arc length of $x = t^2, y = \frac{2}{3}t^3$ from $t = 0$ to $t = 1$.

a. $\frac{2}{3}(2\sqrt{2} - 1)$

b. $\frac{1}{3}(2\sqrt{2} - 1)$

c. $\sqrt{2} - 1$

d. $2\sqrt{2}$

27 [Foundational] • Target: Slopes and Tangents of Polar Curves

Given $r = f(\theta)$, how are the Cartesian coordinates related to polar coordinates?

a. $x = r\cos\theta, y = r\sin\theta$

b. $x = r\sin\theta, y = r\cos\theta$

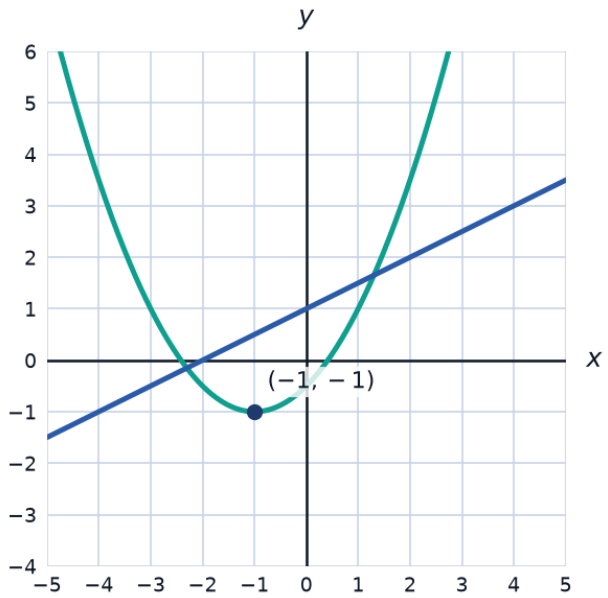
c. $r = x + y, \theta = y/x$

d. $x = r^2\cos\theta, y = r^2\sin\theta$

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28 [Foundational] • Target: Slopes and Tangents of Polar Curves

Find the slope $\frac{dy}{dx}$ of the polar curve $r = 2$ at $\theta = \pi/4$.



a. -1

b. 1

c. 0

d. Undefined

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Slopes and Tangents of Polar Curves

At what angles $\theta \in [0, \pi)$ does the cardioid $r = 1 + \cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \pi$

b. $\theta = \pi/2$

c. $\theta = 0$

d. $\theta = 3\pi/4$

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30 [Intermediate] • Target: Area in Polar Coordinates

State the formula for the area enclosed by a polar curve $r = f(\theta)$ from $\theta = \alpha$ to $\theta = \beta$.

a. $A = \frac{1}{2}\int_{\alpha}^{\beta} [f(\theta)]^2 \, d\theta$

b. $A = \int_{\alpha}^{\beta} f(\theta) \, d\theta$

c. $A = \pi\int_{\alpha}^{\beta} [f(\theta)]^2 \, d\theta$

d. $A = \frac{1}{2}\int_{\alpha}^{\beta} f(\theta) \, d\theta$

31 [Intermediate] • Target: Area in Polar Coordinates

Find the area enclosed by one petal of the four-petaled rose $r = 2\cos(2\theta)$.

a. $\frac{\pi}{2}$

b. π

c. 2π

d. $\frac{\pi}{4}$

32 [Intermediate] • Target: Area in Polar Coordinates

Find the area of the region inside the circle $r = 4\sin\theta$.

a. 4π

b. 16π

c. 2π

d. 8π

33 [Intermediate] • Target: Derivatives of Parametric Curves

Arc length of $r = a\sin^2(\theta/2)$? Recognise this equals $a(1 - \cos\theta)/2$. Total length:

a. $4a$

b. $2a$

c. $8a$

d. πa

34 [Intermediate] • Target: Arc Length of Parametric Curves

Spherical coordinates: arc length element $ds^2 =$

a. $dr^2 + d\theta^2 + d\phi^2$

b. $dr^2 + r^2 d\theta^2 + r^2\sin^2\theta \, d\phi^2$

c. $dr^2 + r^2 d\phi^2$

d. $dr^2 + r^2 d\theta^2$

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35 **[Intermediate]** • **Target: Slopes and Tangents of Polar Curves**

Polar area between two curves $r_1(\theta) \le r_2(\theta)$:

a. $\frac{1}{2}\int (r_2^2 - r_1^2)d\theta$

b. $\frac{1}{2}\int (r_2 - r_1)^2 d\theta$

c. $\int (r_2 - r_1)d\theta$

d. $\pi(r_2^2 - r_1^2)$

36 **[Intermediate]** • **Target: Area in Polar Coordinates**

Consider the parametric curve defined by $x(t) = 3\cos(t) - \cos(3t)$ and $y(t) = 3\sin(t) - \sin(3t)$ for $t \in [0, 2\pi]$. This curve traces out a shape known as a 'cardioid'. Find the maximum area of a triangle that can be inscribed within the loop of this cardioid, with one vertex at the origin.

a. $\frac{27\sqrt{3}}{2}$

b. $\frac{81}{2}$

c. $\frac{81\sqrt{3}}{4}$

d. $54\sqrt{3}$

37 **[Intermediate]** • **Target: Parametric derivatives and arc length**

A rectangular garden is to be enclosed by a fence. One side of the garden is adjacent to a long straight river, and requires no fencing. The other three sides require fencing, and the total length of the fence available is 120 meters. Let the side of the garden parallel to the river have length x meters, and let the two sides perpendicular to the river each have length y meters. The area of the garden is given by $A(x, y) = xy$. \\ Find the dimensions x and y that maximize the area of the garden.

a. x = 60 m, y = 30 m

b. x = 30 m, y = 60 m

c. x = 40 m, y = 40 m

d. x = 80 m, y = 20 m

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Polar area • Cross-Lesson: 7.02 Integration by Parts

A particle's position in the xy-plane is given by the parametric equations $x(t) = 2t + 1$ and $y(t) = t^2 - 3$, where t is time. What is the Cartesian equation of the path traced by the particle?

a. $y = \frac{1}{2}x^2 - 3$

b. $y = \left(\frac{x-1}{2}\right)^2 - 3$

c. $y = x^2 - 3$

d. $y = 2\left(\frac{x-1}{2}\right) + 1$

39 [Challenging] • Target: Second derivatives and concavity for parametric curves • Cross-Lesson: 8.01 Area Between Curves

Epicycloid with n cusps (outer radius R, small circle radius $r = R/n$) has arc length:

a. $8nR$

b. $8R$

c. $4nR$

d. $2\pi nR$

40 [Challenging] • Target: Area between polar curves • Cross-Lesson: 8.02 Volumes of Revolution

Find the length of the curve $x = e^t\cos(t)$, $y = e^t\sin(t)$, $0 \le t \le 2$.

a. $-2\sqrt{2}(1 - e^2)$

b. $-\sqrt{2}(1 - e^2)$

c. $-\frac{\sqrt{2}(1-e^2)}{2}$

d. $1 - \sqrt{2}(1 - e^2)$

41 [Challenging] • Target: Horizontal and vertical tangents on parametric curves • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Arc length of the parametric curve $x = t^2$, $y = t^3$, $0 \le t \le 1$:

a. $\sqrt{13} - 1$

b. $\frac{1}{27}(13\sqrt{13} - 8)$

c. $\frac{2}{27}(13\sqrt{13} - 8)$

d. 1

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42 [Challenging] • Target: Converting between polar and Cartesian for calculus • Cross-Lesson: 9.01 Separable Differential Equations

Find the length of the curve $x = e^t\cos(t)$, $y = e^t\sin(t)$, $0 \le t \le \ln(3)$.

a. $2\sqrt{2}$

b. $4\sqrt{2}$

c. $\sqrt{2}$

d. $1 + 2\sqrt{2}$

43 [Challenging] • Target: Speed and length for parametric motion • Cross-Lesson: 1.01 Precalculus Review for Calculus

Find the length of the curve $x = e^t\cos(t)$, $y = e^t\sin(t)$, $0 \le t \le 1$.

a. $2\sqrt{2}(-1 + e)$

b. $\frac{\sqrt{2}(-1 + e)}{2}$

c. $1 + \sqrt{2}(-1 + e)$

d. $\sqrt{2}(-1 + e)$

44 [Challenging] • Target: Slope and concavity for polar-derived curves • Cross-Lesson: 11.01 Sequences and Series

Find the length of the curve $x = t^2$, $y = \frac{t^3}{3} - t$, $0 \le t \le 4$.

a. $\frac{152}{3}$

b. $\frac{38}{3}$

c. $\frac{76}{3}$

d. $\frac{79}{3}$

45 [Challenging] • Target: Arc length of a polar curve • Cross-Lesson: 12.01 Convergence Tests

Find the length of the curve $x = t^2$, $y = \frac{t^3}{3} - t$, $0 \le t \le 2$.

a. $\frac{28}{3}$

b. $\frac{14}{3}$

c. $\frac{7}{3}$

d. $\frac{17}{3}$

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